

Task 12 – Log Monitoring & Analysis

Tools Used

- Linux authentication logs
- Windows Event Viewer
- Splunk Free

1. Understand Log Types

I learned that logs store system activities.

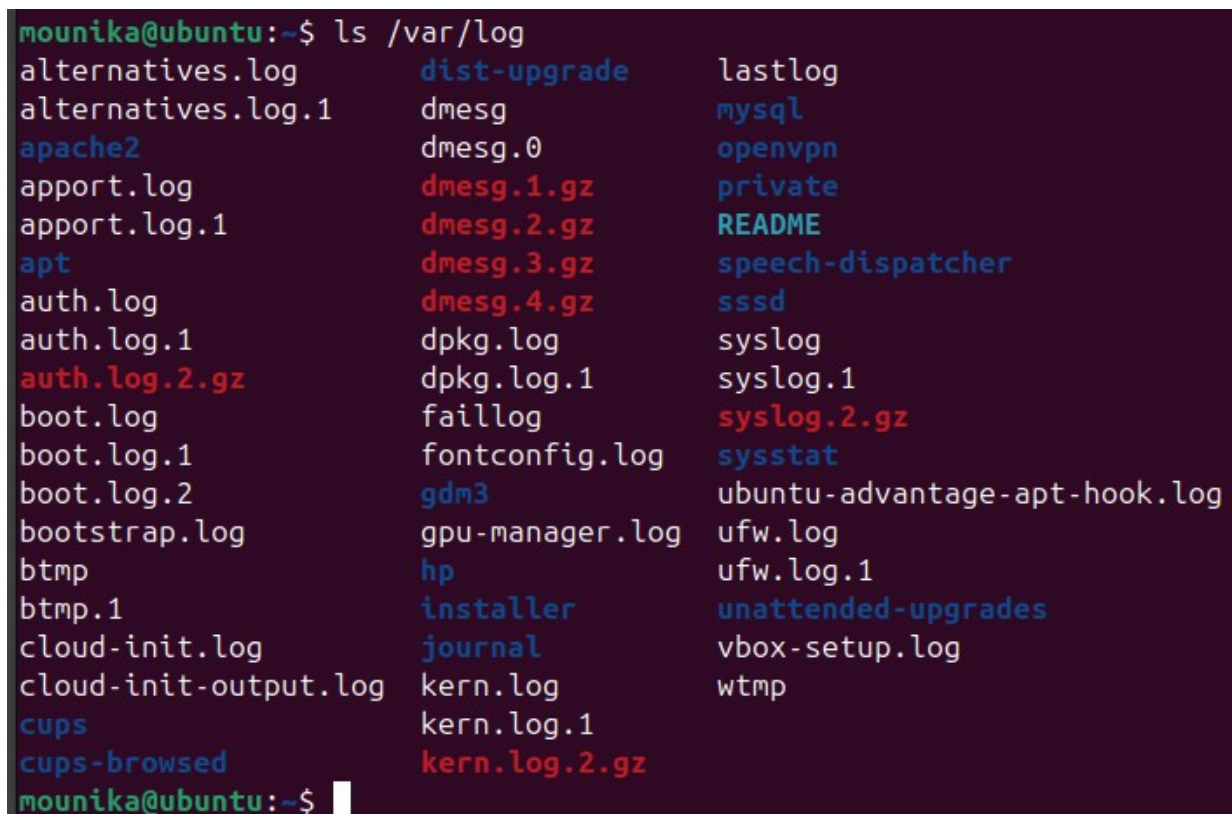
In **Ubuntu**, security logs are inside `/var/log/auth.log`.

In **Windows**, login logs are stored in **Event Viewer** → **Security**.

This showed me that both systems record user activity but in different formats.

Ubuntu

Screenshot : Linux `/var/log` directory

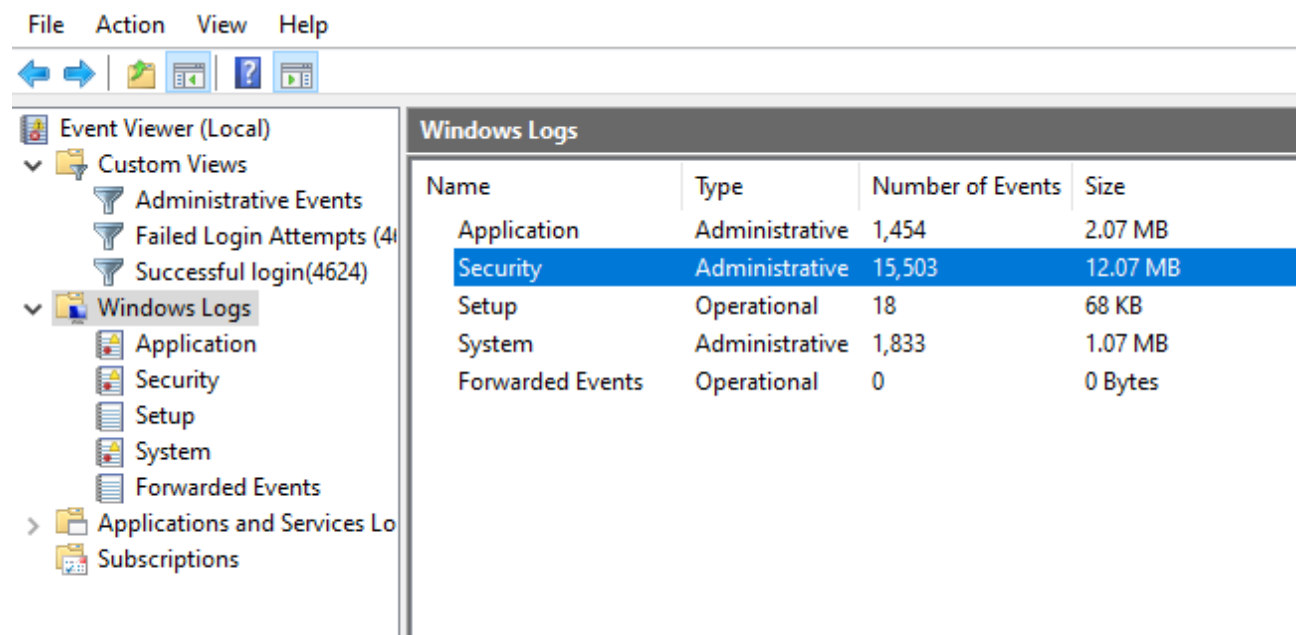


```
mounika@ubuntu:~$ ls /var/log
alternatives.log      dist-upgrade          lastlog
alternatives.log.1    dmesg                 mysql
apache2              dmesg.0              openvpn
appport.log          dmesg.1.gz           private
appport.log.1        dmesg.2.gz           README
apt                 dmesg.3.gz           speech-dispatcher
auth.log             dmesg.4.gz           sssd
auth.log.1           dpkg.log              syslog
auth.log.2.gz        dpkg.log.1            syslog.1
boot.log             faillog               syslog.2.gz
boot.log.1           fontconfig.log        sysstat
boot.log.2           gdm3                 ubuntu-advantage-apt-hook.log
bootstrap.log        gpu-manager.log       ufw.log
btmtp                hp                   ufw.log.1
btmtp.1              installer             unattended-upgrades
cloud-init.log       journal              vbox-setup.log
cloud-init-output.log kern.log              wtmp
cups                 kern.log.1
cups-browsed         kern.log.2.gz
```

`auth.log, syslog`

Windows VM :

Screenshot : Security log window



2. Analyze Authentication Logs

In Ubuntu, I opened `auth.log` to see all login and sudo activities.

In Windows VM, I checked Security logs using Event Viewer.

This helped me understand how user authentication is recorded in both systems.

Ubuntu

```
sudo cat /var/log/auth.log
```

Screenshot : `auth.log` open

3. Identify Failed Logins

Using grep in Ubuntu, I filtered failed and successful logins.

In Windows, I filtered Event IDs **4625** and **4624**.

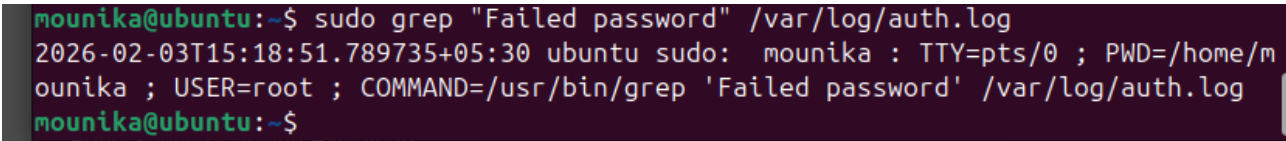
This helped me clearly separate attack attempts and real logins.

Ubuntu

```
sudo grep "Failed password" /var/log/auth.log
```

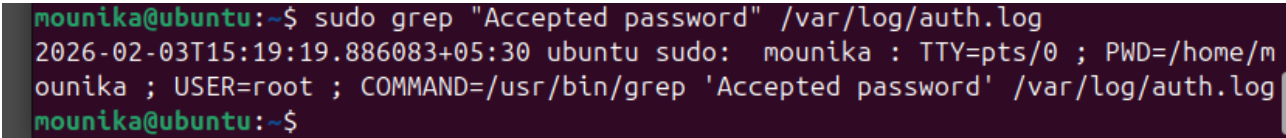
```
sudo grep "Accepted password" /var/log/auth.log
```

Screenshot : Failed logins



```
mounika@ubuntu:~$ sudo grep "Failed password" /var/log/auth.log
2026-02-03T15:18:51.789735+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
mounika@ubuntu:~$
```

Screenshot : Successful logins



```
mounika@ubuntu:~$ sudo grep "Accepted password" /var/log/auth.log
2026-02-03T15:19:19.886083+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/grep 'Accepted password' /var/log/auth.log
mounika@ubuntu:~$
```

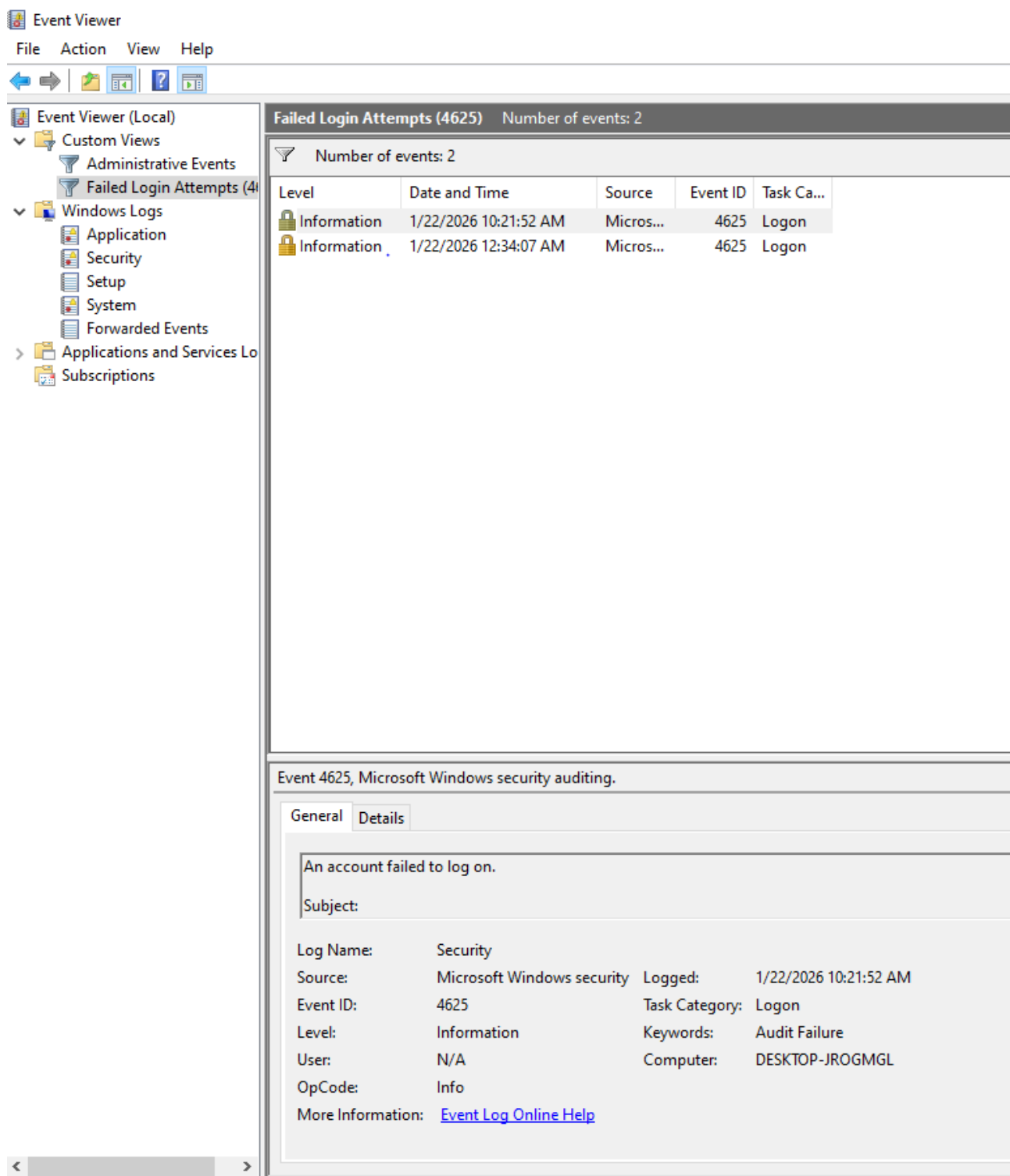
Windows

Filter logs:

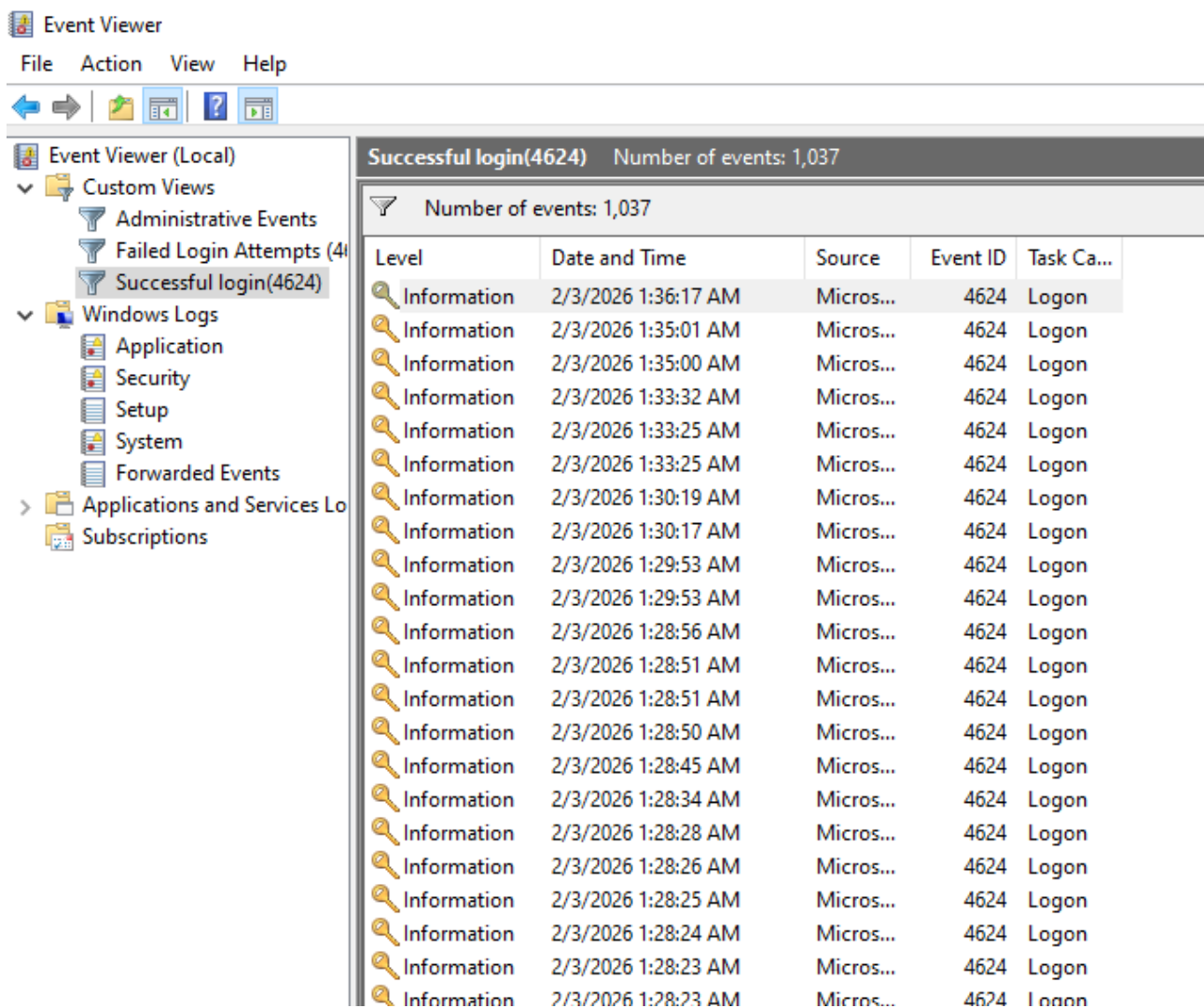
Right click **Security** → **Filter Current Log**

- Event ID: **4625** (failed)
- Event ID: **4624** (success)

Screenshot : Failed login event details



Screenshot : Successful login event details



4. Detect Anomalies

In Ubuntu, I found repeated login failures from the same IP.

In Windows, I saw many failed logins for the same user.

These patterns indicate possible brute force attacks.

Ubuntu

```
sudo grep "Failed password" /var/log/auth.log | awk '{print $1}' | sort | uniq -c | sort -nr
```

Screenshot : Repeated IP attempts

```
mounika@ubuntu:~$ sudo grep "Failed password" /var/log/auth.log | awk '{print $1}' | sort | uniq -c | sort -nr
      2 ;
mounika@ubuntu:~$
```

Windows

Look for many 4625 events from same user/time

Screenshot : Multiple failed logins

New View Number of events: 2				
Number of events: 2				
Level	Date and Time	Source	Event ID	Task Category
Information	1/22/2026 10:21:52 AM	Microsoft Wi...	4625	Logon
Information	1/22/2026 12:34:07 AM	Microsoft Wi...	4625	Logon

5. Correlate Events

In Ubuntu, failed logins followed by sudo commands showed possible compromise.
In Windows, Event ID **4672** showed privilege usage after login.

This correlation proves how attackers move after gaining access.

Ubuntu

`sudo grep "sudo" /var/log/auth.log`

Screenshot : sudo activity

```
mounika@ubuntu:~$ sudo grep "sudo" /var/log/auth.log
2026-02-03T14:52:48.392013+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/cat auth.log
2026-02-03T14:52:48.392488+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-03T14:52:48.402403+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-03T15:15:32.630128+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/cat /var/log/auth.log
2026-02-03T15:15:32.633756+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-03T15:15:32.636317+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-03T15:18:51.789735+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
2026-02-03T15:18:51.791409+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-03T15:18:51.795115+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-03T15:19:19.886083+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/grep 'Accepted password' /var/log/auth.log
2026-02-03T15:19:19.891774+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-03T15:19:19.891774+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-03T15:21:50.679976+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
2026-02-03T15:21:50.681411+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-03T15:21:50.685740+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-03T15:24:17.631511+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/grep sudo /var/log/auth.log
2026-02-03T15:24:17.632945+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
mounika@ubuntu:~$
```

Windows

Filter Event ID **4672**

Screenshot : Privilege assigned event

Event Viewer (Local)

Custom Views

Administrative Events

Failed Login Attempts (4)

Successful login(4624)

New View

Privilege assigned event

Windows Logs

Application

Security

Setup

System

Forwarded Events

Applications and Services Logs

Subscriptions

Privilege assigned event

Number of events: 974

Number of events: 974

Level	Date and Time	Source	Event ID	Task Category
Information	2/3/2026 1:55:07 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:55:04 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:54:30 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:52:00 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:52:00 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:51:54 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:51:52 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:51:51 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:43:16 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:42:38 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:42:37 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:38:59 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:38:52 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:36:17 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:35:01 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:35:00 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:33:32 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:33:25 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:33:25 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:30:19 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:30:17 AM	Microsoft Wi...	4672	Special Logon
Information	2/3/2026 1:29:53 AM	Microsoft Wi...	4672	Special Logon

Event 4672, Microsoft Windows security auditing.

General

Details

Special privileges assigned to new logon.

6. Learn SIEM Basics

From this task, I learned that a **SIEM (Security Information and Event Management)** system is used to collect logs from multiple systems like Linux, Windows, firewalls, and applications into one centralized platform.

Instead of checking each system manually, SIEM tools automatically:

- Collect and normalize logs
- Detect suspicious patterns
- Correlate events
- Generate real-time alerts

Examples of SIEM tools include Splunk, IBM QRadar, and ArcSight.

Even though I did not install a SIEM tool, I understood how it would simplify monitoring by showing all security events in one dashboard and alerting administrators when threats occur.

7. Write Alerts

Trigger	Alert
>5 failed logins	Brute force
Admin login from new IP	Critical
Login at odd hours	Suspicious

8. Document Findings

By documenting logs from both Ubuntu and Windows, I learned how security teams track and stop attacks.

9. Conclusion

This task helped me understand real-world log monitoring and incident detection using both Linux and Windows systems.